

Project Demonstration

Phase 8 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Demonstration Walkthrough

1. Home Page

Displays the 'opti-crop' branding over a satellite farmland background, a short purpose statement, and top navigation links to Home, About, and Find Your Crop.

2. About Page

Presents the full project description: how OptiCrop integrates Nitrogen, Phosphorous, Potassium, soil temperature, humidity, pH, rainfall, and crop types to provide intelligent recommendations that help farmers maximize yields and resource efficiency.

3. Find Your Crop Page

Presents a farmland-themed form titled 'Find Your Best Crop Here' with seven input fields (nitrogen, phosphorous, potassium, temperature, humidity, pH, rainfall) and a Predict button. Submitting valid values displays a message such as 'Best crop for given conditions is coffee' directly beneath the form.

Sample Demonstration Runs

Input (N, P, K, temp, humidity, ph, rainfall)	Recommended Crop
105, 35, 40, 25, 64, 7, 160	coffee
80, 47, 40, 24, 82, 6.4, 236	rice
20, 134, 200, 22.6, 92, 5.9, 113	apple
40, 67, 80, 18.9, 17, 7.3, 80	chickpea

Conclusion

The demonstration confirmed that the trained Logistic Regression model, once deployed through the Flask application, correctly recommends distinct, agronomically sensible crops across a wide range of soil and climate inputs, and that the three-page web interface provides a clear and farmer-friendly

experience from introduction to final recommendation.